

# Natural History





## About the Course

Learners encounter microscopic creatures, pursue the mysterious octopus, and watch the unfolding of life through primary succession. Coordinating afternoon activities are provided in Outdoor Work.

This topic is included in the following course(s): Science: Grade 4

### About Science: Grade 4

In level 4, learners begin to notice science in their community and begin to develop their own interests as they discover a variety of topics from the microscopic world, physical science, and more. A mixture of more and less challenging books is included in this course.

### Science: Grade 4

#### The Big Picture:

To accomplish the goal of supporting a relationship with the Things of the Universe, a Mason science program consists of nature lore, natural history, and general science. Nature immersion, inquiry, community connection, and supportive literature are woven into each of these three parts. Building on the familiarity and foundational knowledge of Form 1, Form 2 learners in grades 4-6 extend the scope of the Things they encounter and begin to notice that science is active in their world. Growing in their self-knowledge at this age, they may notice that they gravitate toward some of these Things more than others. This is a bit like noticing that you have a natural rapport with a particular friend because they seem to understand the way you think and share the same interests. This is wonderful and very typical, but we still want them to get to know more “friends” in science and try different Things, even though one might be their particular interest. Their sense of community is expanding, as well: who are the scientists in their world, and what are they doing? Special care has been taken to build vocabulary and conceptual knowledge “by the way” and to maintain the connection between scientific endeavors and the human experience. In Form 3, students will continue to build on this knowledge of what science is doing in the world when they examine science in its historical context.

Nature lore is timeless knowledge that is passed through a community, much like a grandmother passes on how to make that special bread when the dough just “feels right.” Like Mason, we strive to pass on this knowledge primarily through outdoor work. Group nature walks, seasonal readings, and topics in scouting are provided as an Outdoor Work resource in the Quick Links. If desired, literature suggestions to support lore can be found in the Community Read Alouds resource (in Citizenship Grades 4+).



## Placement & Combining Tips

### About Science: Grade 4

In level 4, learners begin to notice science in their community and begin to develop their own interests as they discover a variety of topics from the microscopic world, physical science, and more. A mixture of more and less challenging books is included in this course.

### Natural History: Grade 4

For approximately fourth-grade students based on a balance of complexity and reading level. However, learners may be freely combined or sequence reordered based on needs and preferences.



## Scheduling

| GRADE | SCHEDULE INFO.                                 | BOOKS                                                                                                                                                           |
|-------|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4     | Natural History: Grade 4<br>1 time/week 15 min | A World in a Drop of Water: Exploring with a Microscope<br>Life on Surtsey: Iceland's Upstart Island<br>The Octopus Scientists: Exploring the Mind of a Mollusk |

### Sample Weekly View

| Day 1                       | Day 2 | Day 3                                                                 | Day 4 | Day 5                                        |
|-----------------------------|-------|-----------------------------------------------------------------------|-------|----------------------------------------------|
| <b>Science: Grade 4</b>     |       |                                                                       |       |                                              |
| Natural History:<br>Grade 4 |       | General Science:<br>Grade 4<br>Nature Walks &<br>Scouting: Grades 1-8 |       | Labs: Grade 4<br>Nature Notebook:<br>Grade 4 |



## Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

**LINKS:** Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

### Science: Grade 4

- ☐ Obtain any supplies indicated on the science or grade-level supply lists.
- ☐ Download any apps and shortcut any desired links.

### Nature Lore:

- ☐ Bookmark your Outdoor Work Quick Link, so that you have it available on your weekly outing. Outdoor Work is generally flexible for your location and season and can be moved around in the schedule to incorporate or substitute Natural History Club outings.
- ☐ Print or bookmark grade-specific nature notebook suggestions to support natural history and general science. Notebooking can be done on a walk, during occupations, or as a field trip, as appropriate.

## Special Topics & Field Trips

### Science: Grade 4

Encourage students to begin choosing their own special topic to observe, study, and research in their field guides. What or who are they curious about or interested in getting to know better? Teachers can choose their own special study, too! Suggested field trips include:

### Natural History: Grade 4

- ☐ Term 1: a pond
- ☐ Term 2: an aquarium
- ☐ Term 3: a controlled burn site

## Term Prep & Teacher Tips

### Science: Grade 4

Object lesson prompts are woven into the lessons, as appropriate, but teachers are encouraged to read corresponding Handbook of Nature Study (HoNS) selections each term and/or any that support special topics chosen by students:

### Natural History: Grade 4

- ☐ Term 1: water lily p.495-497, pondweed p.498-500
- ☐ Term 2: snails/mollusks p.416-421
- ☐ Term 3: one or more insects from p.301-367

## Reminders

Natural History: Grade 4

□ Students will need to collect a sample of pond water for use during Term 1: weeks 1, 8, and 10. There is a reminder to visit a pond in the Nature Notebook section of their Outdoor Work resource, but teachers may want to make a note in their calendars, as well.



## Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.

∞ [View Book List Details](#)

### Natural History: Grade 4



A World in a Drop of Water: Exploring with a Microscope



Life on Surtsey: Iceland's Upstart Island



The Octopus Scientists: Exploring the Mind of a Mollusk



## Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

∞ [View Basic Supplies](#)

∞ [View Supply List Details](#)

### Natural History: Grade 4



Slides and Coverslips



## Quick Links

(No Quick Links Assigned)

Click THIS text  
or scan the QR  
code for links.



# Natural History: Grade 4

## How To Teach



### Introduce

- Begin each lesson with learners' existing knowledge. If the book or activity is new or unfamiliar, then look at the title, a picture, or guidance in the lesson plan to help discuss what students think, drawing on previous experience. If continuing or revisiting a topic or activity, then recap.
- Often, the introduction in the lesson plans or the title of the book's section can help learners to draw out the main idea. Use this to support learners, as appropriate.
- Some learners may benefit from using pictures or looking back briefly.
- Allow them time to share any concerns and come alongside, as needed.



### Read

- Read or do, as instructed in the lessons, noting any Teacher Tips provided. Learners should always have their journal, notebook, or other paper available in case they need to draw or diagram during the lesson.
- Use supportive strategies and educational tools to reduce frustration and better engage the mind, as appropriate. These could include, but are not limited to, the use of eBooks, pictures, audio, read-aloud, buddy reading, colored reading strips, etc.
- If learners do not understand a word or concept, do not worry. Try to show them with a picture or connect the idea to something they have seen in real life. They are learning much by the way and will likely build understanding over the term, the year, and beyond.



### Narrate

- Process the ideas of the lesson by retelling events, describing, explaining a concept, etc. Tips about helping learners deal with various types of passages are provided, but teachers can learn more in Charlotte Mason's School Education Chapter 16, especially p.180.
- Learners may use words, pictures, Legos, PowerPoint, etc, to process and convey ideas in their own way.
- Teachers may take turns to model.



### Discuss

- Consider together any thoughts, confusion, or concerns about the passage, keeping in mind that oftentimes new concepts are not going to be 'mastered' on the first or even second introduction. Mastery is not necessarily the goal, but curiosity and thinking.
- Questions/topics for further discussion are often provided in the lesson plans (or even lab books) to help. There are no right or wrong answers to these. Alternatively, many of these can be used for composition, depending on the needs of the learner and the instructional goals of the teacher. If teachers want to keep ideas active in the mind, these can also be used at other times to keep the ideas in the working memory.
- Note that learners may need to spend more than the allotted time engaging with or even repeating a lesson before moving on or as reinforcement at a later time. Adjust the pace as needed to feed the learner.
- Notice if there were any dates that they want to keep for their Book of Centuries.



## Connect

- Follow any extra links, examine any sidebars in the text, look at pictures, etc., depending on learner needs and interest.
  - These can be viewed as alternative ways to engage.
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# Natural History: Grade 4

[Click THIS text or scan the QR code for links.](#)



## Term 1

### WEEK 1 15m Natural History: Grade 4 - Lesson 1

*A New World*

Materials: A World in a Drop of Water

PREP: Read Teacher Tip

#### → INTRO

Look at the picture on the cover. Have you ever seen such creatures before? In our book, the author will take us on a walk to find such creatures.

#### → READ, NARRATE, & DISCUSS

p.3-5 "Let's go out" - "we have caught."

#### ★ TEACHER TIP

Leave the microscope accessible throughout the term, so that samples can be observed easily in just a few minutes.

#### ★ TEACHER NOTE

Students need to collect samples from a pond this week, as suggested in their nature notebook prompt. These will be used for several weeks DURING lesson activities.

#### ● OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

### WEEK 2 15m Natural History: Grade 4 - Lesson 2

*A Drop of Water Activity*

Materials: microscope, water sample, dropper, slide, cover slip

#### → INTRO

Examine your pond water today.

#### → ACTIVITY

- Make a few slides by transferring a drop of water to a slide and adding a cover slip. It helps to rest the edge of the coverslip on the slide, holding it at an angle. Then allow it to fall onto the drop of water.
- Observe your slide under low power.
- If you find something under low power, try moving up to a higher power objective to see more. Adjust the focus, as needed.

#### → NARRATE & DISCUSS

Use a nature journal to record what the student noticed. When you are finished, let the samples sit away from direct sunlight until you look at them again.

- Living things need food, water, and habitat. Would a living thing in the pond water have all of those?

#### ● OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

### WEEK 3 15m Natural History: Grade 4 - Lesson 3

*Inventing the Microscope*

Materials: A World in a Drop of Water

#### → INTRO

Recall what we read about in the last passage. Let's find out how the microscope was invented before we go further with it.

#### → READ, NARRATE, & DISCUSS

p.6-9 "We are about" - "hundred years ago."

- Why do you think the scientists didn't believe Leeuwenhoek at first? What are some things that can be hard to believe because we can't see them?

#### ● OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

# Natural History: Grade 4

[Click THIS text or scan the QR code for links.](#)



## Term 1

### WEEK 4 15m Natural History: Grade 4 - Lesson 4

#### *Amoeba*

Materials: A World in a Drop of Water, microscope, water sample, dropper, slide, cover slip

PREP: Read Teacher Tip

#### → INTRO

Recall Leeuwenhoek's discovery. Today we'll read about one of his animalcules, the amoeba.

#### → READ, NARRATE, & DISCUSS

p.10-16 "To see what" - "to life again."

#### → ACTIVITY

- Place a drop from your sample of the pond bottom onto a slide and add a coverslip.
- Look at it under the microscope. Is anything alive in there? Did you catch any amoeba?

#### → SUPPLEMENTAL

∞ Video Link: Amoeba (0:47)

#### ★ TEACHER TIP

Supplemental links for optional support are provided at the END of the lesson, but they can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to discussion later. They are NOT required and should NOT take away from the narration and discussion.

#### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

### WEEK 5 15m Natural History: Grade 4 - Lesson 5

#### *Paramecium*

Materials: A World in a Drop of Water, microscope, water sample, dropper, slide, cover slip

#### → INTRO

Tell me what you remember about the amoeba. Today we'll read about another creature, the paramecium.

#### → READ, NARRATE, & DISCUSS

p.17-24 "The amoeba" - "grow older."

#### → ACTIVITY

- Make a slide from your sample of the pond's surface today. Is anything alive in there? Did you catch any paramecia?

#### → SUPPLEMENTAL

It is not necessary to watch the whole video.  
∞ Video Link: Paramecium (5:36)

#### ★ TEACHER NOTE

The video link mentions adaptation and natural selection at 0:56 and 1:51, respectively. Teachers might choose to introduce the terms by the way, mute the narration, or skip this one.

#### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

### WEEK 6 15m Natural History: Grade 4 - Lesson 6

#### *Algae*

Materials: A World in a Drop of Water, microscope, water sample, dropper, slide, cover slip

PREP: Read Teacher Tip

#### → INTRO

Recall what you learned in the last passage. Today we will read about algae, which belong to a very diverse group of living things. Scientists still disagree about how to think about them. In fact, the author refers to algae as plants, but today they are called protists.

#### → READ, NARRATE, & DISCUSS

p.25-30 "Remember the thin" - "on their own."

#### ★ TEACHER TIP

Are your learner(s) demonstrating more ease in using the microscope? More interest/ease in describing the tiny creatures they see with it? If not, consider practicing together more often, exploring extra helpings, or taking a field trip to a museum or lab with microscopes.

#### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

# Natural History: Grade 4

[Click THIS text or scan the QR code for links.](#)



## Term 1

### → ACTIVITY

- Look at your water sample from the surface of the pond. There is probably some algae growing on the top.
- Make a slide from this sample, being sure to capture a bit of this algae. What do you notice?

### WEEK 7 15m Natural History: Grade 4 - Lesson 7

*Euglena*

Materials: A World in a Drop of Water

### → INTRO

Recall what you learned in the last passage. Today we will read about another creature, called euglena.

### → READ, NARRATE, & DISCUSS

p.31-33 "Swimming quickly" - "called protists."

### ★ TEACHER NOTE

Students need to collect samples from a pond again next week, as suggested in their nature notebook prompt.

### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

### WEEK 8 15m Natural History: Grade 4 - Lesson 8

*Rotifer*

Materials: A World in a Drop of Water

PREP: Read Teacher Tip

### → INTRO

Recall what you learned in the last passage. Look at the picture on p.35. The title says this is an animal with wheels. Can you see where the wheels are?

### → READ, NARRATE, & DISCUSS

p.34-37 "All the animals" - "back to life."

- Rotifers are the same size as many of the protists that we met this term, yet they are animals. Why?

### → SUPPLEMENTAL

∞ Video Link: Rotifers Under the Microscope (3:50)

### ★ TEACHER TIP

Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

### WEEK 9 15m Natural History: Grade 4 - Lesson 9

*Hydra*

Materials: A World in a Drop of Water, image link, microscope, water sample, dropper, slide, cover slip, forceps

### → INTRO

Remind me what we read in the last passage. Look at the picture on p.39. Now look at the image of the Greek artifact at the link. Can you see why this animal was named after the monster?

∞ Image Link: Hydra of Greek Mythology

### → READ, NARRATE, & DISCUSS

p.38-44 "Many of" - "into new hydras."

### → ACTIVITY

- Use the forceps to pick up a small piece of your aquatic plant. Examine it to see if you can see anything with the naked eye.
- If it is a thin piece, then place it onto a glass slide and add a cover slip. If it is too thick for the microscope, then use the medicine dropper to rinse the underside off onto the slide with a drop or two of pond water. Then add the cover slip.
- Look closely at your sample under the microscope. What do you see? Is a rotifer there? What about a hydra? Are any of the protists that we

### ★ TEACHER NOTE

The reproduction of hydra is described on p.44: "Hydras can... new hydras." Teachers might allow it to pass by the way, decide if they would like to discuss, or omit the last paragraph.

### ★ TEACHER NOTE

Students need to collect samples from a pond again next week, as suggested in their nature notebook prompt.

### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

# Natural History: Grade 4

[Click THIS text or scan the QR code for links.](#)



## Term 1

previously met there?

### → SUPPLEMENTAL

∞ Video Link: Hydra Under the Microscope (4:25)

#### WEEK 10 15m Natural History: Grade 4 - Lesson 10

*Flatworm*

📄 Materials: A World in a Drop of Water

PREP: Read Teacher Tip

### → INTRO

Remind me what we read in the last passage. Our last tiny creature from the pond is the flatworm.

### → READ, NARRATE, & DISCUSS

p.45-51 "Although we've" - "smaller and smaller."

### → SUPPLEMENTAL

∞ Image Link: Missouri Department of Conservation: Planaria

### ★ TEACHER NOTE

**Planarian sexuality is described on p.48:** "Planarians have... way of reproducing." Teachers might allow it to pass by the way, decide if they would like to discuss, or omit this paragraph.

### ★ TEACHER TIP

The image link will give you an idea of what planaria look like on the underside of rocks. They can be quite small, and their color can be a range of grays and browns.

### ● OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

#### WEEK 11 15m Natural History: Grade 4 - Lesson 11

*Last Look (For Now)*

📄 Materials: A World in a Drop of Water, microscope, water sample, dropper, slide, cover slip

PREP: Read Teacher Tip

### → INTRO

Remind me what we read in the last passage. Today we'll wrap up and then revisit any samples that you want to take one last look at.

### → READ, NARRATE, & DISCUSS

p.52-56 "While we are" - "strange unseen world."

● Use the microscope to revisit any of your samples before you clean out the containers!

### ★ TEACHER TIP

When students take exams next week, evaluate what they recall, but more importantly, whether the microscope or any of the new creatures they have met are more familiar or have possibly even become a particular friend! If no relational growth is observed, consider why: do they need more hands-on time, more reading support, less business, etc?

### ● OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

#### WEEK 12 15m Natural History: Grade 4 - Lesson 12

### → EXAMS

Answer question(s) related to course.

Questions will come from:  
A World in a Drop of Water

### ● OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

# Natural History: Grade 4

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## Term 2

### WEEK 1 15m Natural History: Grade 4 - Lesson 13

*An Adventure to the Pacific*

Materials: The Octopus Scientists

PREP: Read Teacher Tip

#### → INTRO

Look at the picture on the cover. What do you know about octopuses? Octopuses live all over the world, but this book is going to tell us about the search for them on the island of Moorea in the Pacific Ocean. Look at the map on the page following the Table of Contents, noticing the island's location on the lower left. Can you find the location on your globe?

#### → READ, NARRATE, & DISCUSS

p.1-4 "The ocean is" - "eaten themselves."

- What kinds of questions would you ask if you were an octopus scientist?

#### → SUPPLEMENTAL

∞ Image Link: Aquarium of the Pacific: Day Octopus

∞ Video Link: YouTube: Natural History Museum (1:05)

#### → NOTE

We are not going to read the next few paragraphs during lessons ("What good" - "about marine animals."), but if students are interested, feel free to explore this extra passage.

#### ★ TEACHER TIP

This author writes a rich and interesting narrative that weaves together observations about the creature, their ecosystem, the scientists who study them, and the process of science itself. This richness can sometimes feel complex when narrating. If so, teach students to first notice what the purpose of the passage is. Today readers are introduced to the location and the creature that is so difficult to find. How did she begin telling us about that? And then?

#### ★ TEACHER TIP

Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

#### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

### WEEK 2 15m Natural History: Grade 4 - Lesson 14

*Getting Our Feet Wet*

Materials: The Octopus Scientists

PREP: Read Teacher Tip

#### → INTRO

Recall what you read in the last passage. Where has the author taken us, and what are we doing? Do you think they will find an octopus today?

#### → READ, NARRATE, & DISCUSS

p.5-6 "Dripping with" - "Where is it now?"

- Build a food web for the octopus based on what you have read so far.

#### → NOTE

Read "Meet the Octopus Team" p.7-10, or "An Octet of Octo Facts" p.11 for interest, if desired.

#### ★ TEACHER TIP

If students need support with writing or diagramming, teachers should model, scribe, or offer whatever assistance is appropriate.

#### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

# Natural History: Grade 4

[Click THIS text or scan the QR code for links.](#)



## Term 2

### WEEK 3 15m Natural History: Grade 4 - Lesson 15

*On the Hunt*

 Materials: The Octopus Scientists

#### → INTRO

Remind me what we read in the last passage. So, they are on the hunt. Let's see how they are going to find this creature.

#### → READ, NARRATE, & DISCUSS

p.13-16 "What do we" - "anybody yet."

- What would you say is their key strategy to finding the octopus? Is there a creature nearby that you would like to learn more about? If you wanted to look for it using the same strategy as our scientists, how would you go about your search?
- Describe some Thing in nature that you would like to know better.

#### → SUPPLEMENTAL

∞ Image Link: National Geographic: Butterfly Fish

∞ Image Link: National Geographic: Pufferfish

#### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

### WEEK 4 15m Natural History: Grade 4 - Lesson 16

*Hoping for Serendipity*

 Materials: The Octopus Scientists

#### → INTRO

Recall what you read in the last passage. Do you think they will find an octopus at this site?

#### → READ, NARRATE, & DISCUSS

p.16-17 "The afternoon's" - "really good."

- Tell or draw about a time when you were hunting for some Thing in nature.
- Discuss how frustrating it can be when you don't find the creature you are looking for on a walk or when any activity doesn't turn out the way you had hoped. What helps you to persevere?

#### → NOTE

Read "Criobe" p.18 for interest, if desired.

#### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

### WEEK 5 15m Natural History: Grade 4 - Lesson 17

*Continuing the Hunt*

 Materials: The Octopus Scientists

PREP: Read Teacher Tip

#### → INTRO

Recall what you read in the last passage. Let's see how they decide to continue today.

#### → READ, NARRATE, & DISCUSS

p.21-24 "Wet bathing suits" - "a dry towel."

#### → NOTE

Read "How Smart is an Octopus?" p.25 for interest, if desired.

#### ★ TEACHER TIP

Are your learner(s) demonstrating more familiarity with this strange creature? Greater understanding that science is often frustrating? If not, consider discussing big ideas during everyday life, exploring extra helpings, or taking a field trip to an aquarium.

#### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

# Natural History: Grade 4

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## Term 2

### WEEK 6 15m Natural History: Grade 4 - Lesson 18

*Adjusting*

Materials: The Octopus Scientists

PREP: Read Teacher Tip

#### → INTRO

Recall what you read in the last passage. What do you think will happen today?

#### → READ & NARRATE

p.27 "We have been" - "in the water."

#### → READ, NARRATE, & DISCUSS

p.28-31 "The next day" - "hard to study."

- Notice how frustrations and enjoyment sometimes go together. What is something that is frustrating to you? Think of an upside to that activity.

#### → NOTE

Read "Making Friends with an Octopus" p.32 for interest, if desired.

#### ★ TEACHER TIP

Sometimes more complex readings can be broken into smaller parts. In this passage, the first part describes how they plan to go about their study, while the second part tells the story of their first attempt to implement the plan. This is a great opportunity to help students notice that the way we narrate can 'adjust' to the passage!

#### ● OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

### WEEK 7 15m Natural History: Grade 4 - Lesson 19

*Grouping Creatures to Better Understand*

Materials: The Octopus Scientists

PREP: Read Teacher Tip

#### → INTRO

Recall what you read in the last passage.

#### → READ, NARRATE, & DISCUSS

P35-38 "Preserved pufferfish." - "the next day."

- What plants or animals have you noticed that seem to belong to the same family? Even within that family, have you noticed that some seem to be more similar to each other? Try to think of examples, such as some birds eat seeds while others eat bugs or some cats purr while others roar.

#### ★ TEACHER TIP

In this passage, Jennifer is teaching about the concept of classification within the context of the story. If students have difficulty narrating who said what and recalling the finer details, it may help to first notice that larger purpose. Then students can narrate or describe the concept and the process demonstrated by the scientists rather than trying to recall a sequence of events.

#### ● OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

### WEEK 8 15m Natural History: Grade 4 - Lesson 20

*Glimpsing the Octopus*

Materials: The Octopus Scientists

PREP: Read Teacher Tip

#### → INTRO

Recall what you read in the last passage.

#### → READ, NARRATE, & DISCUSS

p.38-41 "For most of" - "so were we!"

- Describe a day in the life of an octopus scientist.

#### → SUPPLEMENTAL

∞ Video Link: YouTube: Octopus Camouflage (1:46)

#### → NOTE

Read "How Octopuses Change Color" p.42-43 for interest, if desired.

#### ★ TEACHER TIP

Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

#### ● OUTDOOR WORK

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# Natural History: Grade 4

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## Term 2

### WEEK 9 15m Natural History: Grade 4 - Lesson 21

*Where They Aren't*

Materials: The Octopus Scientists

#### → INTRO

Recall what you read in the last passage.

#### → READ, NARRATE, & DISCUSS

p.45-49 "We celebrate" - "need to know."

- Science can be a frustrating process! Notice the ways that these scientists overcome their frustration. Can you think of a time when you were able to problem-solve to overcome frustration?

#### → SUPPLEMENTAL

∞ Image Link: Malta National Aquarium: Picasso Triggerfish

#### → NOTE

Read Rainforests of the Ocean p.51-52 for interest, if desired.

#### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

### WEEK 10 15m Natural History: Grade 4 - Lesson 22

*Joy!*

Materials: The Octopus Scientists

#### → INTRO

Recall what you read in the last passage.

#### → READ, NARRATE, & DISCUSS

p.53-57 "By the next" - "Joy!"

- How has their understanding of the geography of the island become important to their hunt? Have you ever noticed creatures that like a certain kind of land feature?

#### → NOTE

Read "Octopus Influence" p.58 for interest, if desired.

#### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

### WEEK 11 15m Natural History: Grade 4 - Lesson 23

*A Lot to Learn*

Materials: The Octopus Scientists

PREP: Read Teacher Tip

#### → INTRO

Recall what you read in the last passage.

#### → READ, NARRATE, & DISCUSS

p.61-65 "The next morning" - "a splendid hostess."

#### → NOTE

Depending on time and student interest, finish conclusion p.65-66 "David named" - "what it's about!" OR just p.66 "In all" - "what it's about!"

#### ★ TEACHER TIP

When students take exams next week, evaluate what they recall, but more importantly, whether the octopus or even the practice of doing science itself has become more comfortable and friendly! If no relational growth is observed, consider why: do they need more hands-on time, more reading support, less business, etc?

#### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

### WEEK 12 15m Natural History: Grade 4 - Lesson 24

#### → EXAMS

Answer question(s) related to course.

Questions will come from:  
The Octopus Scientists

#### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

# Natural History: Grade 4

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## Term 3

### WEEK 1 15m Natural History: Grade 4 - Lesson 25

*Arrival*

Materials: Life on Surtsey

PREP: Read Teacher Tip

#### → INTRO

Look at the picture on the cover. What do you notice about it? Locate Surtsey on the map on p.1. Surtsey is a very young island. This is a story about scientists who are studying its growth and development, but they only get to visit the island for a single week each year.

#### → READ, NARRATE, & DISCUSS

p.1-5 "Drengur Ólafsson" - "for a while."

#### → SUPPLEMENTAL

∞ Video Link: What is a Volcano? (3:14)

#### ★ TEACHER TIP

Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

#### ● OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

### WEEK 2 15m Natural History: Grade 4 - Lesson 26

*The Volcano Erupts*

Materials: Life on Surtsey

#### → INTRO

Recall what you read in the last passage.

#### → READ, NARRATE, & DISCUSS

p.5-6 "From the moment" - "childhood home"

- Can you think of reasons that these scientists would be so determined to study Surtsey?

#### → SUPPLEMENTAL

∞ Video Link: Submarine Volcanoes (3:12)

#### → NOTE

Read "Sizing Up Surtsey" p.7 for interest, if desired.

#### ★ TEACHER NOTE

The supplemental video mentions the age of the Earth at 2:54. Teachers might choose to allow it to pass by the way, consider how they want to discuss with their students, stop the video early, or skip this one.

#### ● OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

### WEEK 3 15m Natural History: Grade 4 - Lesson 27

*To Study Surtsey*

Materials: Life on Surtsey

#### → INTRO

Recall what you read in the last passage.

#### → READ, NARRATE, & DISCUSS

p.10-13 "In 1969" - "Home."

- How do you think it would feel to be chosen to study an isolated, undeveloped island?

#### ● OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

### WEEK 4 15m Natural History: Grade 4 - Lesson 28

*Surtsey Encounters Life*

Materials: Life on Surtsey

#### → INTRO

Recall what you read in the last passage.

#### → READ, NARRATE, & DISCUSS

p.13-17 "The first visitation" - "that never cools."

#### ● OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

# Natural History: Grade 4

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## Term 3

### → NOTE

Read "A Lupine is a Lupine" p.14 for interest, if desired.  
∞ Video Link: Carl Linnaeus's Systema Naturae (3:33)

### WEEK 5 15m Natural History: Grade 4 - Lesson 29

#### *Studying Life*

Materials: Life on Surtsey

PREP: Read Teacher Tip

### → INTRO

Recall what you read in the last passage. Look at the picture on p.18. Can you imagine flying to this island on a helicopter? That's where our team of scientists is going to study life.

### → NOTE

Read the beginning of this chapter for interest, as desired and as time permits.

### → READ, NARRATE, & DISCUSS

p.23-27 "When scientists" - "of the island."

- Why do you think the arrival of the birds changed this part of the island so much?

### ★ TEACHER TIP

Are your learners demonstrating familiarity with the natural world as a system where the plants, animals, and nonliving Things all work together? More interest in volcanos, plants, insects, watching how nature changes? If not, consider observing and discussing together, exploring extra helpings, or taking a field trip to a natural history museum.

### ● OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

### WEEK 6 15m Natural History: Grade 4 - Lesson 30

#### *A Changing Island*

Materials: Life on Surtsey

### → INTRO

Recall what you read in the last passage.

### → READ, NARRATE, & DISCUSS

p.27-31 "Some of the" - "on Surtsey."

- Tell about or draw the meaning of the word 'succession' in natural history.
- Can you explain how each change prepared the island for the next change?

### → NOTE

Read "The 2015 Surtsey Team" p.32 for interest, if desired.

### ● OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

### WEEK 7 15m Natural History: Grade 4 - Lesson 31

#### *The Spit*

Materials: Life on Surtsey

PREP: Read Teacher Tip

### → INTRO

Recall what you read in the last passage. Look at the pictures on p.34, 36 as we walk down to the spit (picture on p.37).

### → NOTE

Read the beginning of this chapter for interest, if desired and as time permits.

### → READ, NARRATE, & DISCUSS

p.37-41 "The spit" - "working with him."

p.44-45 "After a long" - "cooking the dinner!"

### ★ TEACHER TIP

Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

### ★ TEACHER NOTE

The supplemental video mentions evolution at 0:53, 1:02, and 1:06. Teachers might choose to allow it to pass by the way, consider if they would like to discuss

# Natural History: Grade 4

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## Term 3

### → SUPPLEMENTAL

∞ Video Link: Simple Way to Tell Insects Apart (4:06)

alternative ideas with their students, or skip this one.

### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

### WEEK 8 15m Natural History: Grade 4 - Lesson 32

*The Lighthouse*

📄 Materials: Life on Surtsey

### → INTRO

Recall what you read in the last passage. Look at the picture on p.46. Can you find the lighthouse? That's where we're going today.

### → READ, NARRATE, & DISCUSS

p.47-52 "In the wee" - "for a picture."

### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

### WEEK 9 15m Natural History: Grade 4 - Lesson 33

*The Gull Colony*

📄 Materials: Life on Surtsey

### → INTRO

Recall what you read in the last passage.

### → READ, NARRATE, & DISCUSS

p.52-55 "The next stop" - "to a close."

### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

### WEEK 10 15m Natural History: Grade 4 - Lesson 34

*The Last Day*

📄 Materials: Life on Surtsey

### → INTRO

Recall what you read in the last passage.

### → READ, NARRATE, & DISCUSS

p.57-62 "The last full" - "every summer."

### → NOTE

Read "Surtsey's Lava" p.62 for interest, if desired.

∞ Video Link: The Lava Affair (7:53)

### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

### WEEK 11 15m Natural History: Grade 4 - Lesson 35

*Departure*

📄 Materials: Life on Surtsey

PREP: Read Teacher Tip

### → INTRO

Recall what you read in the last passage.

### → READ, NARRATE, & DISCUSS

p.62-65 "He collects" - "this upstart island."

• Explain how new lands sometimes appear in the ocean and then disappear again.

### ★ TEACHER TIP

When students take exams next week, evaluate what they recall, but more importantly, whether the ecosystem or any of the new creatures they have met are more familiar or have possibly even become a particular friend! If no relational growth is observed, consider why: do they need more hands-on time, more reading support, less business, etc?

### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

# Natural History: Grade 4

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## Term 3

### → NOTE

Read Appendices p.66-68 for interest, if desired.

### WEEK 12 15m Natural History: Grade 4 - Lesson 36

### → EXAMS

Answer question(s) related to course.

Questions will come from:  
Life on Surtsey

### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

# Natural History: Grade 4

## Examination

- Term 1** · A World in a Drop of Water: Describe or draw 3 different examples of microscopic pond life.
- Term 2** · The Octopus Scientists: Tell about or draw 3 unique features of the octopus. OR Tell about why this kind of field work is challenging.
- Term 3** · Life on Surtsey: How are the physical conditions, plant life, and animal life interrelated? OR Explain why it is so important for scientists to study Surtsey.